

Bat-Orgil Erdenebat

(662) 380-7252 | batorgilrb@gmail.com | github.com/BatOrgil7 | linkedin.com/in/bat-orgil-erdenebat | Memphis, TN

EDUCATION

University of Mississippi

Oxford, MS

Bachelor of Science in Computer Science – Data Science Emphasis; Math Minor, **GPA: 3.78** Expected May 2027

Relevant Coursework: Algorithms & Data Structure Analysis, Combinatorics, Linear Algebra, Statistics, Machine Learning, Database Systems, Information Visualization, Operating Systems, Computer Organization

Awards: Freeman Grant - \$9,000, Chancellor's Honor Roll Fall 2024, Spring 2025

Certifications: Machine Learning Foundations e-Certificate (Cornell University) Break Through Tech August 2026; Career Essentials in Generative AI (Microsoft, LinkedIn) June 2024

TECHNICAL SKILLS

Languages: Python, R, Java, SQL, C++, Rust, JavaScript, TypeScript, C, LaTeX, Assembly

Libraries/Frameworks: PyTorch/Keras, TensorFlow, Hugging Face Transformers, Scikit-learn, OpenCV, Pandas, NumPy, MediaPipe, Flask, Tailwind CSS, Matplotlib, Seaborn

Tools/Platforms: Linux, Git, GitHub, Neo4j, Google Cloud, MongoDB, MySQL, Docker, Figma, GitLab, AWS

PROFESSIONAL EXPERIENCE

Break Through Tech

New York, NY

AI Studio Fellow

May 2026 – Present

- Selected from **4,000+** applicants nationwide for a one-year AI fellowship focused on machine learning, data science, mentorship, and career readiness.
- Preparing for AI Studio challenge projects with industry advisors to deliver real-world ML solutions and portfolio-ready work.

PotatoNet Co., Ltd.

Seoul, South Korea

Machine Learning Engineering Intern

June 2026 - Present

- Extracted a CNN-LSTM-Attention fault predictor (**94.7%** accuracy) from a legacy Django monolith into a clean FastAPI service on a 16-container microservices platform, and migrated an internal LSTM service to an Nginx → Gunicorn → Django + TLS stack with zero-downtime, reusable deployments.
- Built and shipped a production graph-ETL pipeline that backfilled 6 months of threat data into a live Neo4j store (**19.4K** malicious-JS nodes, 13.7K domains, 3.9K IPs) via idempotent, rollback-tagged loads automated as a nightly self-healing cron job, and deduplicated 267K scripts with SimHash LSH into a leak-free 12.4K-example training set.
- Engineered a 7-signal attacker-profiling engine that clusters malicious JavaScript into named “families” (SimHash AST similarity, perceptual screenshot hashing, and shared TLS-cert/IP/ASN/registrant infrastructure) via GDS Leiden community detection, then diagnosed and fixed a 52K-script over-merge caused by benign library bridges — cutting max family size to **476** across 334 clean families.

University of Mississippi – Dept. of Computer & Information Science

Oxford, MS

Teaching Assistant & Computer Science Tutor

August 2025 – Present

- Led hands-on labs for **200+** STEM students across four courses (Data Structures and Algorithms), lifting Python, Java, and OOP proficiency by coaching students through debugging, optimization, and algorithmic reasoning.
- Raised academic performance for **50+** mentees by delivering individualized tutoring in Python, Java, and Data Structures & Algorithms, reinforcing programming fundamentals and independent problem-solving.

PROJECTS

Census Income Classification — github.com/BatOrgil7/census-income-ml-pipeline

- Built an end-to-end ML pipeline on the U.S. Census Income dataset (32K+ records) to predict annual income. EDA, median imputation, log-transform of skewed features, one-hot encoding, and leakage-free scaling used. Tuned logistic regression via GridSearchCV, scoring on F1 to address class imbalance rather than misleading accuracy.
- Implemented a 3-layer Keras neural network for the same task and found it only matched logistic regression on F1 (0.70 vs 0.68), concluding the simpler, interpretable model was preferable for this tabular problem.
- Audited model coefficients for fairness and surfaced a discrimination risk, informing a recommendation against naive deployment for lending use.

Streaming Evaluation Protocol for CSLR — github.com/BatOrgil7/signstream-eval

- Designed a standardized evaluation protocol for streaming sign language recognition, adapting latency/stability metrics from streaming ASR and simultaneous MT to fill a gap left by offline-only benchmarks.
- Constructed a fault-tolerant landmark extraction pipeline (multiprocessing, content-addressed caching, failure-manifest tracking) processing PHOENIX-2014T across video and frame-folder formats.